

Lemon Fresh Dishwashing Liquid — formulation and plant procedure

Everything needed to make a consistent, market-competitive hand dishwash:
what each chemical does and why, three graded formulas, the exact order of
addition with temperatures and pH targets, the salt curve, quality-control limits,
troubleshooting and safety.

3 grades

10% / 15% / 20% active

11 steps

Batch procedure

10 tests

Release specification

01 — PRINCIPLE

What you are actually making

A hand dishwashing liquid is a water solution of surfactants — molecules with a water-loving head and an oil-loving tail. The tail buries itself in grease, the head stays in the water, and the grease is lifted off the plate as tiny droplets that rinse away. Everything else in the bottle exists to make those surfactants work better, feel better, look better or last longer.

Three numbers decide whether your product beats the one next to it on the shelf:

- **Active matter** — the percentage of real surfactant. This is the single honest measure of quality. Below 10% the customer notices they need more per wash; above 20% you are spending money most buyers will not pay for. Your Rs. 260 / 450ml price point sits comfortably at 15%.
- **Foam quality** — height and, more importantly, how long it survives once grease hits it. Customers judge a dishwash almost entirely by whether the foam dies when the karahi goes in.
- **Viscosity** — 1200–1800 cps. Thin reads as watered-down; too thick pours badly and feels cheap in a squeeze bottle.

Before any production run

Make a 5 kg laboratory batch first, on the exact raw-material lots you will scale up with. Every supplier's LABSA and SLES behave slightly differently, especially on the salt curve. Never make a first 500 kg batch from a formula on paper.

02 — MATERIALS

Raw materials and why each one is there

Buy everything against a Certificate of Analysis and keep the COA with the batch record. Most quality failures in this category start with a cheap drum of something bought without paperwork.

MATERIAL	SPEC TO BUY	JOB IN THE FORMULA
LABSA Linear alkylbenzene sulphonic acid	96% active light colour	The workhorse anionic surfactant and the cheapest source of detergency and foam. It arrives as a dark, thick acid and must be neutralised before it is any use.
Caustic soda Sodium hydroxide flakes	98% min	Neutralises LABSA into its sodium salt. Nothing else in the formula needs it, so the quantity is fixed purely by the LABSA.
SLES Sodium laureth sulphate, 2EO	70% paste	Second anionic. Gives the creamy, stable foam customers associate with a good product, is milder on skin than LAS alone, and is what makes the product respond to salt for thickening.
CAPB Cocamidopropyl betaine	30% active	Amphoteric booster. Raises foam, thickens, and — most usefully — cuts the skin irritation of the two anionics. This is the ingredient that earns the "mild on hands" claim.
CMEA Cocamide MEA (or MIPA)	85–90%	Foam stabiliser. Keeps the lather alive once grease enters the water, and adds body. Prefer MEA or MIPA over the older DEA grade — see the note below.
Glycerine	99.5% USP	Humectant. Leaves a little moisture behind on the skin so hands do not feel stripped after a winter sink-full.
Tetrasodium EDTA	39% solution	Chelating agent. Locks up the calcium and magnesium in hard tube-well water so they cannot kill your foam or cloud the bottle. Genuinely important across most of Pakistan.
Sodium chloride	refined, non- iodised	Viscosity builder. Cheap and effective, but the most dangerous ingredient in the formula for a new operator — read the salt-curve section before you touch it.
Preservative CMIT/MIT or BIT blend	as supplied	Stops bacteria and mould growing in a warm warehouse. A dishwash at neutral pH with no preservative will go cloudy and smell sour within months.
Fragrance	detergent grade	Lemon. Buy a grade specified for anionic detergent systems — perfumes formulated for cosmetics often discolour or drop out here.
Colour	food / D&C grade	Tartrazine for lemon-yellow, sunset yellow for orange, or a blend for green apple. Always dose as a pre-made 1% solution, never as raw powder.
Citric acid	food grade	Final pH trim only. Used as a 20% solution, drop by drop.
Water	soft or DM < 50 ppm hardness	Two-thirds of the bottle. If your water is hard, your product will be cloudy and low-foaming no matter how good the rest of the formula is. Install a softener before you install anything else.

Avoid cocamide DEA

Diethanolamide grades can form nitrosamines and are restricted or banned in several export markets. If you ever want to sell outside Pakistan — or simply want a cleaner ingredient list — use cocamide MEA or MIPA instead. Performance is close to identical at the same dose.

03 — FORMULAS

Three grades, one process

All three run through exactly the same procedure. Only the quantities change. Percentages are by weight (w/w); a 100 kg batch means 100 kg on the scale, not 100 litres.

Master formulation table

INGREDIENT	ECONOMY 10% ACTIVE	STANDARD 15% ACTIVE	PREMIUM 20% ACTIVE
Water (soft/DM)	79.62	70.52	61.57
LABSA 96%	6.00	9.00	11.00
Caustic soda flakes 98%	0.77	1.15	1.41
SLES 70%	5.50	8.00	12.50
CAPB 30%	3.50	6.00	8.00
Cocamide MEA	1.20	1.50	1.80
Glycerine	0.50	1.00	1.50
Tetrasodium EDTA 39%	0.25	0.30	0.35
Sodium chloride (to viscosity)	~2.20	~2.00	~1.20
Preservative	0.08	0.08	0.10
Fragrance	0.25	0.30	0.40
Colour, 1% solution	0.05	0.05	0.05
Citric acid (pH trim)	0.08	0.10	0.12
Total	100.00	100.00	100.00

What each grade is for

GRADE	ANIONIC ACTIVE	USE IT FOR
Economy 10%	10.0%	The 250ml price-fighter and bulk 5L institutional can, where the buyer is choosing on rupees per litre.
Standard 15%	14.8%	Your 450ml Lemon Fresh at Rs. 260. The right balance of performance and cost for the main shelf line.
Premium 20%	20.0%	A concentrated variant you can price above the market and back with a "half the squeeze" claim. Also the base for a hotel/restaurant grade.

The caustic rule of thumb

Caustic soda flakes at 98% =

12.8% of the LABSA weight

. For 9 kg LABSA that is 1.15 kg NaOH. Always confirm against the acid value on your LABSA COA — if it differs from ~165 mg KOH/g, recalculate as: $\text{NaOH (kg)} = \text{LABSA (kg)} \times \text{acid value} \times 0.000728$.

Batch calculator

Pick a grade, type your batch size, and take these numbers straight to the weighing floor. Enter your own purchase rates in the right-hand column and the cost per bottle updates with them.

Batch sheet

Standard grade · 100 kg

INGREDIENT	%	QUANTITY	RATE RS./KG	COST RS.
Water (soft/DM)	70.52	70.5 kg	2	Rs. 141
LABSA 96%	9.00	9.00 kg	340	Rs. 3,060
Caustic soda 98%	1.15	1.15 kg	190	Rs. 218
SLES 70%	8.00	8.00 kg	330	Rs. 2,640
CAPB 30%	6.00	6.00 kg	320	Rs. 1,920
Cocamide MEA	1.50	1.50 kg	620	Rs. 930
Glycerine	1.00	1.00 kg	450	Rs. 450
EDTA 39%	0.30	300 g	420	Rs. 126
Sodium chloride	2.00	2.00 kg	35	Rs. 70
Preservative	0.08	80 g	1900	Rs. 152
Fragrance	0.30	300 g	3500	Rs. 1,050
Colour 1% soln	0.05	50 g	250	Rs. 13
Citric acid	0.10	100 g	400	Rs. 40
Total	100.00	100 kg		Rs. 10,810

PACKAGING & CONVERSION, PER 450ML BOTTLE (RS.)

BOTTLE + CAP

24

LABEL

3

CARTON SHARE

5

LABOUR + POWER

5

FREIGHT

5

WHAT YOU EARN PER BOTTLE, BY CHANNEL

CHANNEL	DISCOUNT	YOU RECEIVE	YOUR COST	PROFIT	MARGIN
Direct retail / own shop	0%	Rs . 260	Rs . 92	Rs . 168	65%
Retailer	18%	Rs . 213	Rs . 92	Rs . 121	57%
Sub-Distributor	26%	Rs . 192	Rs . 92	Rs . 100	52%
District Distributor	34%	Rs . 172	Rs . 92	Rs . 79	46%

BATCH	BOTTLES	PROFIT @ RETAILER	PROFIT @ SUB-DIST.	PROFIT @ DISTRICT
10 litres	22	Rs . 2,660	Rs . 2,203	Rs . 1,745
20 litres	44	Rs . 5,321	Rs . 4,405	Rs . 3,490
50 litres	111	Rs . 13,422	Rs . 11,113	Rs . 8,805
100 litres	222	Rs . 26,844	Rs . 22,227	Rs . 17,609
Per bottle	1	Rs . 121	Rs . 100	Rs . 79

Liquid cost only in the top table. Add bottle, cap, label, carton, shrink film, labour, electricity and freight before you judge the margin — on a 450ml pack those usually add Rs. 30–55 depending on your bottle supplier. Density is taken as 1.03 kg/litre.

05 — BATCH SHEETS

Ready batch sheets — 10, 20, 50 and 100 litres

These are the four sizes you will actually use: 10 litres for a trial, 20 for a pilot, 50 and 100 for routine production. Quantities are by **weight**, because that is how you must weigh them — the litre figure is what comes out at the other end. Density is taken as 1.03 kg per litre, so a 100-litre batch weighs 103 kg.

INGREDIENT	%	10 LITRES	20 LITRES	50 LITRES	100 LITRES
Water (soft/DM)	70.52	7.26 kg	14.5 kg	36.3 kg	72.6 kg
LABSA 96%	9.00	927 g	1.85 kg	4.63 kg	9.27 kg
Caustic soda 98%	1.15	118 g	237 g	592 g	1.18 kg
SLES 70%	8.00	824 g	1.65 kg	4.12 kg	8.24 kg
CAPB 30%	6.00	618 g	1.24 kg	3.09 kg	6.18 kg
Cocamide MEA	1.50	155 g	309 g	773 g	1.54 kg
Glycerine	1.00	103 g	206 g	515 g	1.03 kg
EDTA 39%	0.30	31 g	62 g	155 g	309 g
Sodium chloride	2.00	206 g	412 g	1.03 kg	2.06 kg
Preservative	0.08	8.2 g	16 g	41 g	82 g
Fragrance	0.30	31 g	62 g	155 g	309 g
Colour 1% soln	0.05	5.2 g	10 g	26 g	52 g
Citric acid	0.10	10 g	21 g	52 g	103 g
Batch weight	100.00	10.3 kg	20.6 kg	51.5 kg	103.0 kg
450ml bottles	—	22	44	111	222

Weigh, do not measure by volume

LABSA, SLES and cocamide are thick and their density changes with temperature. A jug will lie to you; a scale will not. For anything under 100 g — preservative, colour, citric acid — use a 0.1 g jeweller’s scale, not a kitchen scale.

How the four sizes differ in practice

BATCH	WEIGHT	450ML BOTTLES	WHAT IT IS FOR
10 litres	10.3 kg	22	Trial batch. Always make this first on a new raw-material lot, and run the full salt curve on it so you know where the peak sits before you commit real money.
20 litres	20.6 kg	44	Pilot. Enough to give samples to two or three distributors and still keep retained bottles for the stability test.
50 litres	51.5 kg	111	Small production run. Comfortable in a 100-litre vessel — never fill a vessel past about two-thirds or you cannot stir without splashing.
100 litres	103 kg	222	Routine batch. Needs a 150–200 litre vessel with a proper anchor stirrer. Below this size a motorised stirrer is optional; at this size it is not.

Scaling the caustic solution

The caustic is always dissolved in its own water first, at 30% strength, taken out of the batch water. For a 100-litre batch that is 1.18 kg of flakes in 2.8 kg of water. Subtract that water from the first charge, not from the final make-up, or your batch will finish over weight.

06 — PROCEDURE

Step-by-step batch procedure

Written for a 100 kg Standard batch in a stainless-steel 316 or HDPE vessel with an anchor stirrer at 60–90 rpm. Never use a mild-steel tank: the acid stage and the caustic stage will both attack it and your product will pick up iron and turn grey.

1

Prepare and check

Calibrate the pH meter with buffer 4 and buffer 7. Confirm the water is soft. Weigh out everything in advance and keep each container labelled — the worst batch failures come from adding the right material at the wrong moment.

Water < 50 ppm hardness

pH meter calibrated

2

Make the caustic solution — do this first

In a separate HDPE drum, take 3.8 kg of the batch water — enough to make a 30% solution. Add the 1.15 kg of caustic flakes **to the water**, slowly, stirring throughout. The drum will get hot — 50 to 60 °C is normal. Set it aside and let it cool below 35 °C. This takes 30–45 minutes, which is why it goes first.

30% solution

NaOH → water, never the reverse

Cool to < 35 °C

3

Charge water and LABSA

Put 45 kg of water in the main vessel and start the stirrer. Add the 9 kg of LABSA slowly over 10–15 minutes. It is dark and thick and will look wrong — that is expected. Keep stirring until it is evenly dispersed.

45 kg water

9.00 kg LABSA

10–15 min

4

Neutralise — the step that decides the batch

Add the cooled caustic solution slowly over 20–30 minutes, checking pH every few minutes. The mix will lighten from dark brown to pale amber and go clear as it neutralises. Stop at pH 7.0–7.5 and hold the temperature below 45 °C.

Do not overshoot. Above pH 8.5 you will hydrolyse the SLES you are about to add, the fragrance will go off, and the product will yellow on the shelf. If you do overshoot, bring it back with citric acid solution before going any further.

Target pH 7.0–7.5

Max 45 °C

Then stir 15 min

5

Cool before the SLES goes in

Let the batch fall below 35 °C. SLES hydrolyses in heat, which shows up weeks later as a cloudy bottle that smells faintly of alcohol and has lost its foam. This wait is not optional.

< 35 °C

6

Add the secondary surfactants

Add 8 kg SLES 70% in portions — it is a stiff paste and dumping it in creates lumps that take an hour to dissolve. Keep the stirrer blade below the surface so you are not whipping air into the batch. Stir until completely clear, 20–30 minutes.

Then 5 kg CAPB, stir 10 minutes. Then 1.5 kg cocamide MEA — if it is a waxy flake, pre-melt it or dissolve it in a little warm water first. Stir until there are no undissolved specks.

8.00 kg SLES

5.00 kg CAPB

1.50 kg CMEA

7

Additives

1.00 kg glycerine, stir 5 minutes. 0.30 kg EDTA solution, stir 5 minutes. 0.08 kg preservative, stir 10 minutes.

Order here is not critical, but add each one fully before starting the next.

8

Fragrance and colour — cool batch only

Below 32 °C, pour the pre-made 1% colour solution in and stir until even. Then add 0.30 kg fragrance slowly into the vortex, not down the vessel wall. Stir 15 minutes. Adding perfume to a warm batch simply evaporates money.

< 32 °C

Colour as 1% solution

9

Trim the pH

Check pH again. If it has drifted above 7.5, bring it back with 20% citric acid solution added drop-wise. Final target for a hand dishwash is 6.5–7.5 — neutral is what keeps it mild.

10

Build the viscosity with salt

This is the step that ruins batches. Never add dry salt. Make a 25% brine, add it in small increments, and read the salt-curve section in full before you start.

25% brine only

0.2% steps

See salt curve

11

Make up, rest, release

Add the remaining water to bring the batch to exactly 100 kg and stir 20 minutes. Then let it stand 8–12 hours so entrained air escapes and the product goes bright. Run the full QC panel, and only then filter through a 100-mesh screen and fill. Print the batch number on every bottle.

Rest 8–12 h

100-mesh filter

Batch code on pack

07 — CRITICAL**The salt curve, and why it destroys batches**

Salt thickens a sulphate-based surfactant system by changing the shape of the micelles. As you add it, viscosity climbs — slowly at first, then steeply. It reaches a peak. And then, with barely any more salt, it falls off a cliff and the batch goes watery.

The cruel part is that the fall is not reversible. Once you are past the peak, adding more salt makes it thinner, and adding water makes it thinner still. The only fix is to blend the ruined batch into a fresh un-salted one, which costs you a day.

How to do it safely

- Make a **25% brine**: 25 kg salt dissolved in 75 kg water. Never add dry salt to the batch — it dissolves unevenly and you cannot control the dose.
- Add in steps of **0.2% of batch weight as salt** — for a 100 kg batch that is 0.8 kg of brine per step.
- After each step, stir 10 minutes, draw a sample, cool it to 25 °C, and measure viscosity. Room-temperature readings on a warm batch will lie to you.
- Write every reading down and plot it. After three or four batches you will know your own curve and can go straight to a starting dose.
- **Stop on the rising side**, at 1200–1800 cps. Do not chase the peak. You want to be comfortably below it so that normal batch-to-batch variation cannot push you over.

If the batch has gone past the peak

Stop adding anything. Set it aside, make a fresh batch with no salt at all, and blend the two — usually 1 part ruined to 2 parts fresh — then re-titrate the salt slowly on the combined lot. Do not try to rescue it with thickener or by boiling off water.

Thickening without salt

If you find your curve is uncomfortably sharp, raise the CAPB by 1% and the cocamide MEA by 0.3% instead. Both thicken gently and forgivingly, and both improve the product. They cost more than salt, which is exactly why salt is used — but a ruined batch costs more than either.

08 — QUALITY

Release specification

No batch leaves the plant until it passes every line of this table. Keep a retained sample of every batch for two years and file the results with the batch record — when a distributor complains, the retained sample is what settles the argument.

TEST	METHOD	SPECIFICATION
Appearance	Visual, 25 °C, against white	Clear and bright. No haze, no sediment, no undissolved specks.
Colour	Visual against retained standard	Matches the approved standard. Note any drift.
Odour	Smell test, two people	Clean lemon. No sour, soapy, oily or solvent note.
pH (as is)	pH meter, 25 °C	6.5 – 8.0 target 7.0 – 7.5
Active matter	Hyamine 1622 two-phase titration	≥ 15.0% (Standard grade)
Viscosity	Brookfield RVT, spindle 2, 20 rpm, 25 °C	1200 – 1800 cps
Specific gravity	Hydrometer, 25 °C	1.020 – 1.050
Foam	Ross-Miles, 0.1% solution, 25 °C	≥ 150 mm initial ≥ 130 mm at 5 min
Cold stability	5 °C for 24 h, then warm to 25 °C	No haze, no crystals, returns fully clear.
Heat stability	45 °C for 4 weeks (accelerated ≈ 1 year)	No separation. Viscosity drift ≤ 10%. Colour and odour hold.

The cheapest test you can run

Keep a greasy-plate test as a routine: a measured smear of used cooking oil on a standard plate, a fixed dose of product in a fixed volume of 40 °C water, and count how many plates one sink-full will clean before the foam dies. It correlates with what your customer actually experiences far better than any number in the table above, and it costs nothing.

09 – DIAGNOSIS

Troubleshooting

SYMPTOM	MOST LIKELY CAUSE	WHAT TO DO
Batch too thin	Not enough salt, or active matter below target	Add brine in 0.2% steps. If it will not thicken at all, check active matter — you may have under-charged LABSA or SLES.
Batch too thick or gelled	Overshot the salt peak, or CMEA over-dosed	Blend into a fresh un-salted batch at roughly 1:2 and re-titrate.
Cloudy or hazy	Hard water, un-neutralised LABSA, or cold storage	Check pH first. If pH is fine, it is the water — switch to softened or DM. For winter clarity, add 1–2% urea.
Goes cloudy only in winter	Cloud point too high	Add 1–2% urea, or raise CAPB slightly. Confirm with the 5 °C cold-stability test.
Sour smell after a few weeks	Microbial growth, or SLES hydrolysis	If pH held at 6.5–7.5, it is microbial: raise the preservative and check your water and vessel hygiene. If pH drifted low, the SLES broke down.
Yellowing on the shelf	Over-neutralised, poor-colour LABSA, or sunlight	Hold pH under 7.5, buy a lighter-colour LABSA grade, and store cartons out of direct sun.
Foam dies on greasy dishes	Active matter too low, or not enough foam stabiliser	Raise cocamide MEA by 0.3% and CAPB by 1%. If it is still weak, the active matter is the real problem.
Colour fades in the bottle	Light-sensitive dye	Switch to a light-stable dye at the same shade, or move to a pigmented bottle. Keep a retained sample in the dark to confirm.
Separation into layers	Incomplete dissolution, or an incompatible addition	Usually CMEA that never fully dissolved. Extend stirring and pre-melt it next time.
Skin complaints from users	pH drift high, or CAPB under-dosed	Verify pH is 6.5–7.5 on the retained sample. Raise CAPB and glycerine.

10 — IMPORTANT

The antibacterial claim — read before you print labels

There is a trap here that catches a lot of new manufacturers. The usual antibacterial agents sold locally — benzalkonium chloride and similar quaternary ammonium compounds — are **cationic**, positively charged. Your formula is built on **anionic**, negatively charged surfactants. Put them together and they neutralise each other: the quat stops killing bacteria, the surfactants stop foaming, and you may get a cloudy precipitate as well.

So you cannot simply stir a quat into this formula and call it antibacterial. You have two honest options:

- **Claim removal, not killing.** Surfactants physically lift bacteria off the surface along with the grease, and the rinse carries them away. Wording like "removes bacteria from dishes and utensils" is accurate for this formula and is what most major brands now use. This is what the Sofex label and website already say.
- **Reformulate around a compatible system** if you genuinely want a kill claim. That means moving to a non-ionic and amphoteric base so a quat can survive, which costs more and foams noticeably less — and it means you should back the claim with a laboratory test report, not an assumption.

Do not print a kill claim you cannot evidence

A "kills 99.9% of germs" line on a pack needs a test report behind it. If a regulator or a competitor asks and you have nothing, the cost is your brand, not a fine. Claim what the product actually does — it is a strong enough story on its own.

11 — COMMERCIAL**Cost, selling price and profit**

Three quarters of your liquid cost sits in three drums. Knowing which ones tells you where to spend your negotiating effort — a Rs. 20/kg saving on LABSA is worth more than anything you can do to the salt or the colour.

Where the money goes

Cost share of the liquid, at the rates entered in the calculator. Change a rate there and this table follows.

MATERIAL	RATE RS. /KG	KG PER 100 KG	COST	SHARE
LABSA 96%	340	9.00	Rs. 3,060	28.3%
SLES 70%	330	8.00	Rs. 2,640	24.4%
CAPB 30%	320	6.00	Rs. 1,920	17.8%
Fragrance	3,500	0.30	Rs. 1,050	9.7%
Cocamide MEA	620	1.50	Rs. 930	8.6%
Glycerine	450	1.00	Rs. 450	4.2%
Caustic soda 98%	190	1.15	Rs. 218	2.0%
Preservative	1,900	0.08	Rs. 152	1.4%
Water (soft/DM)	2	70.52	Rs. 141	1.3%
EDTA 39%	420	0.30	Rs. 126	1.2%
Sodium chloride	35	2.00	Rs. 70	0.6%
Citric acid	400	0.10	Rs. 40	0.4%
Colour 1% soln	250	0.05	Rs. 13	0.1%
Liquid, per 100 kg		100.00	Rs. 10,810	100%

Cost per 450ml bottle

COST LINE	INDICATIVE	NOTE
Liquid	Rs . 50	Standard grade at the indicative chemical rates. Moves most with LABSA and SLES.
Bottle + flip-top cap	Rs . 24	Your biggest non-liquid line. Buy in truckloads and negotiate hard — a rupee saved here is a rupee earned on every single bottle.
Label	Rs . 3	Falls sharply at higher print runs.
Carton, partition, tape	Rs . 5	Per bottle, spread over a carton of 12.
Filling labour + power	Rs . 5	Drops once you move from hand filling to a semi-auto filler.
Freight to distributor	Rs . 5	Depends on distance and load size.
Total factory cost	Rs . 92	This is the number every selling price has to clear.

Selling price and profit, by channel

At Rs. 260 MRP and Rs. 92 total cost. The live version of this table, driven by your own rates, sits in the calculator.

CHANNEL	DISCOUNT	YOU RECEIVE	PROFIT	MARGIN
Direct retail / own shop	0%	Rs . 260	Rs . 168	65%
Retailer	18%	Rs . 213	Rs . 121	57%
Sub-Distributor	26%	Rs . 192	Rs . 100	52%
District Distributor	34%	Rs . 172	Rs . 80	46%

Profit per batch

BATCH	BOTTLES	LIQUID COST	PROFIT @ RETAILER	PROFIT @ DISTRICT
10 litres	22	Rs . 1,114	Rs . 2,660	Rs . 1,745
20 litres	44	Rs . 2,227	Rs . 5,321	Rs . 3,490
50 litres	111	Rs . 5,568	Rs . 13,422	Rs . 8,805
100 litres	222	Rs . 11,135	Rs . 26,844	Rs . 17,609

These rates are estimates, not quotes

Industrial chemical prices in Pakistan are not published — they are quoted per order and move with the rupee and with crude. Every figure above is an indicative starting point. Get written quotes from at least three suppliers for LABSA, SLES and CAPB, enter your real numbers in the calculator, and work from those. Do not agree a distributor discount before you have done this.

Check the right number

At Rs. 260 MRP a district distributor at 34% off pays you Rs. 172, not Rs. 260. That Rs. 172 is what your Rs. 92 cost has to sit under — and it still has to cover your salary, rent, electricity, wastage and bad debt. A 46% gross margin at the deepest tier is healthy; anything under about 25% means that tier is not worth serving.

12 — SAFETY

Safety on the floor

Two of your raw materials cause serious chemical burns. Treat them accordingly, every single time, including on the thousandth batch.

Caustic soda

- Always add caustic **to water**. Adding water to caustic flakes can boil violently and throw hot alkali at the operator's face.
- Dissolving generates real heat — 50–60 °C is normal. Use an HDPE drum, not glass, and never seal it.
- Full-face shield over goggles, elbow-length nitrile gloves, rubber apron, closed shoes.
- Alkali burns are painless at first and keep damaging tissue for minutes. Flush with running water for at least 20 minutes and go to hospital — do not judge by how it feels.

LABSA

- A strong acid. Same PPE as caustic. Store the drums away from the caustic store.
- The neutralisation step is exothermic. Add the caustic solution slowly and watch the thermometer — a runaway exotherm can boil and splash the batch.

General

- Plumbed eyewash station and a safety shower within ten seconds' walk of the mixing vessel. Test them weekly.
- Cross-ventilation over the mixing area.
- **Never store or mix any of this near hypochlorite bleach.** If you also make bleach, keep it in a separate room with separate equipment and separate staff clothing.
- Keep a Material Safety Data Sheet for every raw material in a folder on the floor, not in an office drawer, and make sure the operators have actually read the first page.
- Label every intermediate container. An unlabelled drum of caustic solution is how people get hurt.

13 — COMPLIANCE

Labelling and compliance

Confirm the current requirements with PSQCA directly before you finalise artwork — standards are revised and the list of items requiring a quality mark changes. The items below are the ones that are consistently expected on a household cleaning product sold in Pakistan.

- Product name and function — "Dishwashing Liquid"
- Net content in millilitres, declared honestly and verified by your fill checks
- Manufacturer's full name and address — Sofex Soap & Chemical Industries, with your registered address
- Batch number, manufacturing date and best-before or expiry date
- List of ingredients
- Directions for use
- Caution statements — keep out of reach of children, avoid contact with eyes, rinse thoroughly if swallowed, do not mix with bleach
- Maximum retail price, if you print MRP on pack
- Your NTN / sales tax registration, per FBR requirements for your business
- PSQCA quality mark, if this product category is on the compulsory certification list at the time you print

Worth doing early

Get one batch independently tested at a recognised laboratory for active matter, pH and viscosity, and keep the report. It costs very little, it settles distributor disputes instantly, and it is the first thing a serious buyer or a modern-trade chain will ask you for.

Sofex Soap & Chemical Industries — internal production manual.

These are starting formulations, not a substitute for your own validation. Raw materials vary between suppliers and lots, so every formula here must be confirmed with a laboratory-scale trial batch and a full stability study on your own materials before it goes into production. Verify current PSQCA specifications and labelling requirements directly with the authority before printing packaging.